Sparse edge cancellation in unknot recognition: complete vertex-disc search, explicit obstructions, and compressed boundary interfaces

A research continuation for Vladimir Reshetnikov's ProveIt project

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Abstract

We develop a parameter for normal-disc discovery that is independent of the number of occupied quadrilateral types. For an oriented normal meridian disc, its geometric edge weights differ from the absolute values of its signed intersection cocycle only on edges carrying cancellation. On a one-vertex triangulation with first Betti number one, a meridian whose cancellation set has size k lies in a known linear space of dimension at most $k+1$. We prove that all standard vertex meridians with this property can be recovered by small exact linear systems, without first discovering their quadrilateral sectors. A sparse-incidence refinement reduces the number of systems from an elementary $t^{2k+O(1)}$ bound to $t^{k+O(1)}$ times a factor depending only on k. A supported matching minor and boundary cohomology certify the promised vertex discs in polynomial bit time, without component-orbit queries. We extend the discovery theorem to multiple vertices using sign chambers of the cocycle space.

The new search produces a 47-piece vertex disc on the maintained eight-tetrahedron example for which the entire primitive coherent family has genus two. Its cancellation is confined to one interior edge. An explicit one-dimensional calculation explains this disc, and boundary layering gives a growing family with the same cancellation parameter. Conversely, parallel Whitehead curves give finite triangulations of solid tori on which every meridian has linearly many cancellation edges; this obstruction allows multiple vertices. We also recover cyclic attachment data for compressed parallelity bases and distinguish the correct relative handle replacements for different disc-base interfaces.

The package contains additive code, independent certificate replay, controlled experiments, and proofs with explicit hypotheses. The parameterized discovery theorem is unconditional. A logarithmic parameter bound on a constructively chosen family of knot exteriors is not proved, so a general quasi-polynomial unknot recognizer is not claimed.

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1 Research question and precise scope

The maintained project already has substantial exact infrastructure: normal-coordinate validation, compressed component and disc observers, independently checked diagram exteriors, coherent cocycle searches, complete enumeration inside supplied sectors of small dimension, bounded Pachner searches, and compressed cut geometry. Its most recent source also contains an all-dimensional source-parallelity construction with a linear bound on the inherited complementary core [1]. The question addressed here is how to discover useful normal disc vectors without guessing all quadrilateral types or expanding their possibly enormous integer coordinates.

The 2021 lecture announcement is the motivation for the quasi-polynomial target [5]. The July 2026 paper gives a new hierarchy algorithm and polynomially checkable hierarchy certificates. Its Section 9 explicitly leaves some step costs unspecified; Proposition 9.1 bounds iterations by $L(g+1)^L$, and the refined construction controls hierarchy length linearly [4]. These facts do not supply a quasi-polynomial bound for the maintained implementation. Our results are a separate, parameterized route to discovery.

1.1 What is established

The main results are the following.

1. A complete exact search for standard vertex meridian discs with at most k cancellation edges, using spaces of dimension at most $k+1$ on one-vertex, rank-one sources (Theorem 4.2).
2. A sparse-incidence enumeration of the determining equations with an explicit $t^{k+O(1)}$ system-count bound (Theorem 5.1). This is a stronger quantitative statement than multiplying the number of edge subsets by the number of equation subsets.
3. An extension with parameter $p+k$, where $p = \dim H^1(M;\mathbb{R}) + v - 1$ is the dimension of the cocycle space (Theorem 6.1).
4. A new explicit disc and an infinite layered family with cancellation parameter exactly one; and, in a different class allowing multiple vertices, a linear lower bound for every meridian.
5. Recovery of the cyclic boundary attachments of compressed parallelity bases, together with the appropriate relative handle descriptions for disc bases.

None of the first three statements assumes that a useful quadrilateral sector has already been supplied. None proves a logarithmic bound on the new parameter for a universally available source family. The last distinction is the remaining geometric issue, not a hidden linear-algebra cost.

1.2 Meaning of novelty and validation

Normal edge coordinates, cocycle-dual surfaces, vertex normal surfaces, and hyperplane-based ray enumeration are established tools. Edge-weight coordinates are explicitly exposed by Regina [9]. The local four-point calculation below is included with proof, rather than claimed as a new fact about normal surfaces. The proposed contribution is their combination into a complete cancellation-parameter search, its sparse enumeration bound, its explicit source families, and the added compressed interface operator. A literature check and comparison with the pinned repository found no earlier statement of this particular package of results. This is not a claim of exhaustive priority verification.

Computations are used to discover examples and test implementations. Finite tests do not establish asymptotic complexity or replace the proofs. Likewise, a positive certificate for a supplied triangulation does not certify that it is the exterior of a particular input diagram. That source correspondence remains an explicit input to any recognition-level use.

2 Normal surfaces as compatible four-point metrics

Let $\mathcal{T}$ be a finite triangulation of a compact orientable 3-manifold. Self-identifications between distinct faces are permitted, but local edge occurrences and their orientations are retained. Write t for the number of tetrahedra and m for the number of global edges; always $m \le 6t$. For a normal surface S, let

$$w_e(S) = |S \cap e| \quad (e \in E(\mathcal{T}))$$

be its geometric edge weights. Coordinates count intersections, not signed intersection numbers and not the sum over repeated local occurrences of a global edge.

In one tetrahedron label the vertices $0, 1, 2, 3$. Write t_i for the four triangle coordinates and $q_{ij|kl}$ for the quadrilateral separating $\{i, j\}$ from $\{k, l\}$. This quadrilateral avoids the opposite edges ij, kl. At most one of the three quadrilateral coordinates is positive. Define the three opposite-edge sums

$$s_0 = w_{01} + w_{23}, \qquad s_1 = w_{02} + w_{13}, \qquad s_2 = w_{03} + w_{12}.$$

For every face and every one of its corners put

$$a_i^{jk}(w) = \frac{w_{ij} + w_{ik} - w_{jk}}{2}. \tag{1}$$

These are the numbers of normal arcs cutting off the corresponding corner.

Lemma 2.1 (Local reconstruction)**.** *Nonnegative real edge weights in a tetrahedron come from a unique admissible real standard normal vector if and only if all twelve corner arc values in (1) are nonnegative and the maximum of s_0, s_1, s_2 is attained at least twice. If all three sums agree, all quadrilateral coordinates are zero. Otherwise the strictly smaller sum identifies the avoided pair of edges, and its quadrilateral coordinate is half the difference between the larger and smaller sums. In either case*

$$t_i = \min_{\{j,k\} \subset \{0,1,2,3\} \setminus \{i\}} a_i^{jk}(w). \tag{2}$$

For integral edge weights, the reconstructed coordinates are integral exactly when every face has even total edge weight.

Proof. If $q = q_{01|23}$ is the occupied type and $T = t_0 + t_1 + t_2 + t_3$, the edge equations give

$$s_0 = T, \qquad s_1 = s_2 = T + 2q.$$

The other types give the other two permutations of this relation. At a fixed vertex, two of the corner arc values equal t_i, and the third equals $t_i + q$. This proves necessity and the reconstruction formula.

Conversely suppose, for example, $s_0 \le s_1 = s_2$. Set $q = (s_1 - s_0)/2$ and use (2) for the triangles. Direct substitution into the six edge equations gives

$$w_{01} = t_0 + t_1, \quad w_{23} = t_2 + t_3, \quad w_{ij} = t_i + t_j + q \quad (ij \notin \{01, 23\}).$$

The assumed corner inequalities give $t_i \ge 0$, and the sum inequality gives $q \ge 0$. The formulas determine every coordinate, proving uniqueness. The all-equal case is the same calculation with $q = 0$.

For integer weights, the parity condition makes every number in (1) an integer. It also makes the differences of opposite sums even: for example $s_1 - s_0$ is even by subtracting the parity equations on the faces 012 and 123. Thus all reconstructed coordinates are integral. The converse follows because each normal arc contributes two endpoints to a face. □

The calculation can be pictured as the distance matrix of a tree with four labelled leaves. The pendant lengths are t_i, and the single internal length is q. This picture includes zero lengths and coincident leaf locations; strict positivity is not required.

Proposition 2.2 (Global edge-coordinate reconstruction)**.** *An assignment of nonnegative real weights to the global edges of $\mathcal{T}$ is an admissible real normal vector exactly when the local conditions of Lemma 2.1 hold in every tetrahedron. Local reconstruction then gives its unique standard coordinates. If the weights are rational, clearing the denominators of those coordinates gives a genuine normal surface.*

Proof. On a paired face the three global edge weights are the same on both sides, with their local labels permuted. Formula (1) therefore gives the same three normal arc counts. These are precisely the normal matching equations. Conversely a normal surface restricts to each tetrahedron as in the lemma. Unique reconstruction on every tetrahedron gives global uniqueness. Multiplying rational coordinates by a common denominator preserves all matching, nonnegativity and quadrilateral constraints. □

2.1 One common set of hyperplanes

Fix a compatible choice Q of at most one allowed quadrilateral type in each tetrahedron. The image $\mathcal{C}_Q$ of its standard normal cone in edge coordinates is a pointed rational polyhedral cone. The edge map is a linear isomorphism on this cone and its linear span: express any vector in the span as a difference of cone vectors and apply local uniqueness. Its defining equalities and inequalities have their boundary hyperplanes among:

1. the m equations $w_e = 0$;
2. the three equations $s_i - s_j = 0$ per tetrahedron;
3. the twelve equations $2a_i^{jk}(w) = 0$ per tetrahedron.

Denote this list by $\mathcal{H}$, and write

$$H = m + 15t \leq 21t. \tag{3}$$

Repeated or identically zero equations may be removed. Each row has support on at most four global edges, including after identifications; its coefficients have uniformly bounded absolute value. To describe a particular $\mathcal{C}_Q$, one imposes the appropriate equality between the two larger sums, the inequality for the third sum, and the corner inequalities. This description explains why searching the common hyperplane list does not require prior knowledge of Q.

3 Signed intersections and cancellation support

Give every global edge an orientation. For a cooriented properly embedded normal surface S, let $c_e(S)$ be the algebraic intersection number with that edge. A face contains oriented normal arcs with cancelling endpoint signs, so its oriented edge sum is zero. Thus $c(S)$ is an integral 1-cocycle. If an edge has p_e positive and n_e negative intersections, then

$$w_e = p_e + n_e = |p_e - n_e| + 2\min(p_e, n_e) = |c_e| + 2z_e, \qquad z_e \in \mathbb{Z}_{\geq 0}. \tag{4}$$

Definition 3.1. The cancellation support of the cooriented surface is $F(S) = \{e : z_e > 0\}$. Its size counts edges with cancellation, not the number of cancelling pairs. Cancellation multiplicities may be very large. For a triangulated solid torus, $\kappa_{\mathrm{v}}(\mathcal{T})$ is the minimum of $|F(D)|$ over standard vertex normal meridian discs D.

By a standard vertex surface we mean that its coordinate ray is an extreme ray of the full nonnegative standard matching cone and its coordinates are primitive integers. For an admissible

vector this is equivalent to extremality in any containing admissible coordinate sector: such a sector is a coordinate face of the matching cone. The usual vertex-disc existence theorem makes the set in the definition nonempty for a triangulated solid torus [2]. We keep the word *vertex* in the parameter because the discovery proof below uses extremality. It does not show that an arbitrary small-cancellation disc can be replaced by a vertex disc with equally small cancellation.

Lemma 3.2 (One vertex fixes the cocycle)**.** *Suppose M is connected, orientable, and has torus boundary, $\mathcal{T}$ has one global vertex, and $b_1(M) = 1$. Let c be a primitive integer generator of its cocycle lattice. An oriented essential disc with boundary on the torus boundary has intersection cocycle c or $-c$. Consequently its weight vector satisfies*

$$w(D) = b + 2z, \qquad b = (|c_e|)_{e \in E}. \tag{5}$$

Proof. All 0-coboundaries vanish when there is one global vertex. The cocycle lattice is therefore $H^1(M; \mathbb{Z})$, a free abelian group of rank one. An essential simple closed curve on a torus is nonseparating. A small boundary collar therefore supplies a path from one side of the disc to the other without recrossing it; equivalently the properly embedded disc is nonseparating. A closed path crossing it once shows that its dual cohomology class is primitive. Its cocycle is consequently one of the two generators. Equation (5) now follows from (4). □

The absolute vector b is itself an admissible normal vector in edge coordinates. On a tetrahedron a cocycle is the difference of four local heights, so $b_{ij} = |h_i - h_j|$. These are collinear four-point distances and satisfy Lemma 2.1. The associated surface is the familiar coherent cocycle-dual surface. It need not be a disc.

Lemma 3.3 (The coherent positive support has small rank)**.** *Let C be the nonzero coherent surface of an integral cocycle c. The standard matching kernel restricted to its positive support has dimension at most $p = \dim Z^1(\mathcal{T}; \mathbb{R})$. In particular, when $p = 1$ and c is primitive, C is a primitive standard vertex surface and is connected, although its genus need not be zero.*

Proof. Fix an ordering of the original cocycle heights in each tetrahedron. The supported coordinates are its nonzero successive height gaps: a triangle at the minimum, the middle quadrilateral, and a triangle at the maximum. A tied height constrains its corresponding gap to zero. For any vector y in the supported matching kernel, regard these coordinates as formal height gaps. On each local oriented edge the formal difference equals $\operatorname{sign}(c_e) w_e(y)$, with the local-to-global edge orientation included. This identity is linear and does not require y to be nonnegative. If $c_e = 0$, all supported normal types avoid that edge, so $w_e(y) = 0$.

Matching makes these edge weights agree at all occurrences of a global edge. The vector $a_e = \operatorname{sign}(c_e) w_e(y)$ is consequently a global cocycle. The map $y \mapsto a$ is injective: zero a implies zero weights, and the edge map is injective on the fixed quadrilateral support. The dimension bound follows. If $p = 1$, the nonzero vector C forces dimension one. A common coordinate divisor of C divides every weight $|c_e|$, so primitive c makes C primitive. The primitive-ray component argument of Proposition 4.4 proves connectedness. □

3.1 Computing the primitive generator

Build the integral face matrix A, using one signed equation per global triangular face and summing repeated edge occurrences. Rank-one nullspace computation and division by the coordinate gcd produce c. The one-vertex and rank-one hypotheses are checked, not inferred from a filename. Each row of A has sum of absolute coefficients at most three. Cofactors of an independent $(m-1)$-row submatrix therefore give a generator with

$$\max_e |c_e| \le 3^{m-1}. \tag{6}$$

Hadamard's inequality proves this bound, and primitive division only decreases it. Exact computation has polynomial bit complexity. In particular, exponential geometric weights do not entail exponential input length for this stage.

4 Complete discovery from low-dimensional weight spaces

For an edge subset F define

$$W_F = \operatorname{span}_{\mathbb{R}}(\{b\} \cup \{\mathbf{e}_f : f \in F\}), \qquad d_F = \dim W_F \le |F| + 1. \tag{7}$$

Here $\mathbf{e}_f$ is the coordinate vector of edge f. We deliberately search the whole space W_F. Requiring a nonnegative cancellation vector would make a smaller search, but is unnecessary for soundness: every returned candidate is reconstructed and topologically checked. In particular, membership in W_F alone is not an orientation certificate.

Lemma 4.1 (Determining an extreme ray by active equations)**.** *Let $W \subset \mathbb{R}^m$ be a d-dimensional linear space and let $w \neq 0$ generate an extreme ray of a sector cone $\mathcal{C}_Q$. If $w \in W$, then some $d-1$ equations from $\mathcal{H}$, restricted to W, are independent and have common kernel $\mathbb{R}w$ in W.*

Proof. The ray stays extreme in $\mathcal{C}_Q \cap W$: a nonproportional nonnegative decomposition there would also be a decomposition in $\mathcal{C}_Q$. Consider all defining equalities and all inequalities active at w. If their common kernel in W had dimension greater than one, choose a direction u not parallel to w in that kernel. For sufficiently small $\varepsilon > 0$, both $w + \varepsilon u$ and $w - \varepsilon u$ satisfy every inactive strict inequality as well as the active equations. They give a nonproportional decomposition of $2w$, a contradiction. Thus the active kernel is precisely the line $\mathbb{R}w$. Select an independent $(d-1)$-row subset. Every selected hyperplane belongs to $\mathcal{H}$ by (3). The argument includes sectors and intersections that are not full dimensional. □

Theorem 4.2 (Complete cancellation-supported vertex-disc discovery)**.** *For a supplied finite triangulation of a compact connected orientable 3-manifold with exactly one torus boundary, one global vertex and $b_1 = 1$, there is a deterministic exact algorithm that finds an essential disc whenever a standard vertex essential disc has at most k cancellation edges. The algorithm makes at most*

$$\sum_{\substack{F \subseteq E \\ |F| \le \min(k,m)}} \binom{H}{d_F - 1} \le \sum_{j=0}^{\min(k,m)} \binom{m}{j}\binom{H}{j} \tag{8}$$

small homogeneous linear-system attempts. Each candidate has polynomially bounded encoded length and is checked in polynomial bit time. Its overall bit cost is $t^{O(k+1)}$. If the input is only known to satisfy the source hypotheses, exhaustion of a smaller k is inconclusive for unknot recognition.

Proof. For every F with $|F| \le k$, compute a basis of W_F. Enumerate all $(d_F - 1)$-subsets of $\mathcal{H}$ and retain those of restricted rank $d_F - 1$. Find the one-dimensional rational kernel, test its two directions for nonnegativity, and apply Lemma 2.1 on every tetrahedron. Clear denominators in the reconstructed standard coordinates, divide by their gcd, and apply the polynomial vertex-disc gate of Proposition 4.4. The zero-dimensional case contributes no ray; a one-dimensional space needs the empty equation subset.

If D is a promised vertex disc, take its cancellation set F. Equations (5) and (7) put its edge ray in W_F. The edge isomorphism preserves its extremality. Lemma 4.1 ensures that one enumerated system recovers exactly this ray. Primitive normalization then recovers D. The tests cannot manufacture a positive result: they verify genuine normal coordinates and essential disc topology in the supplied manifold.

The bound counts all systems before pruning or duplicate removal. Exact linear algebra, reconstruction and the vertex-disc gate are polynomial in the source and coordinate bit lengths. The bit estimate below makes these lengths polynomial in t, k and the size of c. This proves the stated parameterized bound without an assumption about the run time of the repository's present interval-orbit engine. □

Remark 4.3 (The displayed count). The second inequality in (8) is used for $j \le H/2$, which includes the relevant range $j \le m \le 6t$ since $H \ge m + 15t$. Then $d_F - 1 \le j$ and the binomial coefficients are increasing on that range. Zero hyperplanes can be left on the list for this upper bound.

4.1 Encoded size and verification cost

Let B bound the bit length of b_e and of the source indices. A basis of W_F can be selected from b and the coordinate vectors. Restricted hyperplane coefficients have $O(B+1)$ bits. For a d-column system, cofactor bounds give a null vector whose coordinates have

$$O(d(B + \log(d+1))) \tag{9}$$

bits. Multiplication by the weight-space basis and the half-integer local reconstruction preserve a polynomial bound. The produced standard coordinates may describe exponentially many discs, but their binary encoding and all verification inputs remain polynomial in t, k, B. No estimate is measured in the expanded piece count.

4.2 A polynomial gate for the promised vertex discs

There is a useful way to avoid a stronger-than-needed topology subroutine. Let M_N be the standard normal matching matrix, and for a nonzero integral admissible vector x let $J = \operatorname{supp} x$. Its submatrix $M_{N,J}$ retains exactly the positive coordinate columns.

Proposition 4.4 (Rank and boundary parity certify a vertex disc)**.** *Suppose ∂M is a torus. A primitive nonzero integral admissible normal vector x is an essential disc if all three tests hold:*

$$\dim \ker M_{N,J} = 1, \qquad \chi(x) = 1, \qquad [\beta(x)] \neq 0 \text{ in } H^1(\partial M; \mathbb{F}_2),$$

where $\beta(x)_e = w_e(x) \bmod 2$ on boundary edges. Every primitive standard vertex meridian passes these tests. All tests, including admissibility, primitivity and the boundary coboundary quotient, have polynomial bit complexity in the encoded source and x.

Proof. The nullity test says exactly that the positive support of x spans a ray of the standard matching cone. If the surface were disconnected, each component would have an integral normal vector $y_i = \lambda_i x$ with $\lambda_i > 0$ and $\sum_i \lambda_i = 1$. Since $\gcd(x) = 1$, there are integers a_j with $\sum_j a_j x_j = 1$. Thus $\lambda_i = \sum_j a_j (y_i)_j$ is an integer. There can be only one component.

On every boundary face, the sum of the three intersection parities is zero: each normal arc has two endpoints. Hence β is a cocycle. It is the mod-two Poincaré dual of the boundary curve system, because its evaluation on a transverse edge counts precisely those intersections. The nonzero class implies that the connected surface has nonempty boundary. The classification of compact connected surfaces now makes $\chi = 1$ equivalent to being a disc. Its single boundary curve is essential on the torus exactly when its mod-two homology class is nonzero: an essential simple torus curve has primitive integral slope.

Conversely, a standard vertex disc has a one-dimensional supported kernel, Euler characteristic one, and an essential simple boundary with nonzero mod-two class. Matching ranks are exact rational linear algebra; Euler characteristic is a linear normal-coordinate calculation; and testing

whether β belongs to the image of the boundary vertex coboundary map is linear algebra over $\mathbb{F}_2$. All matrices have $O(t)$ rows and columns. Determinant bounds control the rational bit lengths. □

The parity must be computed *after* primitive standard-coordinate normalization. A doubled disc has zero boundary parity. If the boundary has several vertices, a nonzero raw edge-parity vector may still be a coboundary, so quotienting by vertex potentials is necessary.

This gate is sufficient for the entire vertex-disc discovery theorem. It may reject additional rays that are extreme only in the chosen weight space; the completeness promise never required those rays. The main search prototype instead reuses the maintained source-bound disc-component observer and its independent replay, which can accept more surfaces. That implementation has a bound in its realized interval traces; it is not identified here with the abstract polynomial AHT theorem [3]. A separate exact implementation of the rank gate accompanies the package. Positive total Euler characteristic alone is never a positive certificate: it can be contaminated by spheres or inessential components.

5 Sparse incidence removes one combinatorial factor

The elementary search chooses F and then chooses equations. Most such pairs cannot have the required rank. Each equation in $\mathcal{H}$ involves at most four edges, so an invertible determining system must have a matching between its rows and the chosen defect edges. This observation provides a different enumeration order.

Theorem 5.1 (Incidence enumeration)**.** *The completeness conclusion of Theorem 4.2 can be achieved with at most*

$$1 + \sum_{j=1}^{\min(k,m)} \left(4^j \binom{H}{j} + m\, 4^{j-1} \binom{H}{j-1} \right) \tag{10}$$

exact system attempts, up to duplicate descriptions. Preparation and work per system are polynomial in t, k, B. Thus the bit bound has the form $t^{k+O(1)} \exp(O(k \log(k+1))) \operatorname{poly}(B)$, with a fixed polynomial exponent outside k. The second summand handles the exceptional case in which all nonzero coordinates of b belong to the cancellation set.

Proof. First suppose F has size j and $\operatorname{supp} b \not\subset F$. The listed basis $b, \mathbf{e}_f$ is independent. A meridian has weight $b + u$, with $\operatorname{supp} u \subset F$, so its coefficient of b is one. Choose j independent active equations as in Lemma 4.1. Write their coefficient matrix in these basis coordinates as $(a \mid R_F)$. The square matrix R_F is invertible: otherwise a nonzero vector supported on F would lie in their common kernel, whereas that kernel is the meridian line and every nonzero point on that line has a nonzero b-coefficient.

A nonzero term in the determinant of R_F matches each row to a different edge in its support. Enumerate all j-subsets of the global hyperplane list and all selections of one supported edge per row. There are at most 4^j selections. Reject repeated edges and singular resulting matrices. For each surviving edge set solve

$$R_F u_F = -a. \tag{11}$$

Reconstruct and check $w = b + u$. The active system for every promised meridian is among these attempts. The method may also return harmless extra surfaces; all positive outputs receive the same topological check.

Now suppose $\operatorname{supp} b \subset F$. The meridian weight is supported on F itself. In the j-dimensional coordinate space $\mathbb{R}^F$, its ray is determined by $j - 1$ independent active equations. Normalize by $\sum_{f \in F} w_f = 1$. The resulting $j \times j$ matrix is invertible, since the active kernel is the line spanned by the nonnegative meridian vector and the normalization is nonzero on that ray. A nonzero

determinant term matches the normalization row to some one of the m edges and matches every other row to one of at most four supported edges. Enumerating the $j-1$ hyperplanes, these choices, and the associated normalized systems therefore includes every such ray. This gives the second summand in (10). Cases with actual cancellation size below k are covered at their own value of j.

All matrices have order at most $k+1$ and polynomially bounded entries. The same cofactor and verification analysis applies. Substituting $H \le 21t$ into (10), and bounding the finite sum, gives the displayed complexity form. The factor depending only on k absorbs integer arithmetic and elementary combinatorial bookkeeping of each attempt. □

Primitive normalization may rescale an affine candidate $b+u$. Therefore the enumerated set F is a search-space description, not automatically the actual cancellation support of a returned disc. To certify that parameter, recompute its final edge weights and compare them with b. This does not affect completeness: a promised disc occurs at its correct affine scale before primitive normalization.

5.1 The radius-one specialization

For $k=1$ the nonexceptional candidates are particularly simple. A row $\ell \in \mathcal{H}$ and an edge $e \in \operatorname{supp} \ell$ give

$$u_e = -\frac{\ell(b)}{\ell_e}. \tag{12}$$

There are at most $4H$ row-edge incidences. Equivalent rational changes can be deduplicated before reconstruction. The exceptional sparse rays are coordinate rays. This avoids rebuilding the complete hyperplane list for every possible defect edge.

Equation (12) gives a linear number of small arithmetic candidates, not an unconditional linear-time recognizer. A candidate can affect every tetrahedron incident to a high-degree edge, and the complete vertex-disc gate still has its own polynomial cost. A correct implementation reports both the number of determining systems and the actual complete query time.

6 Several vertices and the quasi-polynomial implication

With several vertices, fixing a cohomology class does not fix its cocycle. Integral vertex potentials add coboundaries. For a connected triangulation with v vertices, put

$$p = \dim Z^1(\mathcal{T};\mathbb{R}) = v - 1 + b_1(M).$$

Let C be an $m \times p$ rational basis matrix for its cocycle space. Every signed intersection cocycle is $C\alpha$. A fixed-class version uses a cohomology generator together with a coboundary basis; for a knot exterior these again have total dimension v.

Theorem 6.1 (Sign-chamber extension)**.** *For a finite triangulation of a compact connected orientable 3-manifold with torus boundary and $p \ge 1$, standard vertex essential discs with at most k cancellation edges can be discovered in*

$$t^{O(p+k)} \operatorname{poly}(B)$$

bit time from a supplied triangulation. Here B bounds the encoded rational basis and source. For a knot exterior $p = v$.

Proof. Consider the central arrangement in $\mathbb{R}^p$ with equations $(C\alpha)_e = 0$, omitting rows that vanish identically. Enumerate its full-dimensional sign chambers. Their number is at most

$$2\sum_{i=0}^{p-1}\binom{m-1}{i}. \tag{13}$$

The standard recurrence for the number of regions proves this: adding one hyperplane creates at most as many regions as its induced arrangement has in dimension $p-1$. Degeneracies only decrease the number. An exact incremental enumeration using rational strict-feasibility tests has $m^{O(p)}\,\mathrm{poly}(B)$ bit cost. Strict homogeneous inequalities may be replaced by lower bounds of one when seeking a representative of a nonempty chamber, since a representative can be positively rescaled.

Give each identically zero row the arbitrary sign $+1$. For a chamber sign vector σ, let D_σ be the corresponding diagonal sign matrix. Every point in its closure satisfies $|C\alpha| = D_\sigma C\alpha$, including coordinates that vanish. Every cocycle lies in at least one such closure, so no separate enumeration of lower-dimensional sign cells is necessary.

For each chamber and each $|F| \le k$, use

$$W_{\sigma,F} = \mathrm{im}(D_\sigma C) + \mathrm{span}_{\mathbb{R}}\{\mathbf{e}_f : f \in F\}.$$

This space has dimension at most $p+k$. Equation (4) puts every promised disc in one of these spaces. Apply Lemma 4.1 and the reconstruction procedure to its at most $p+k-1$ determining hyperplanes. Multiplying the chamber count, the edge-subset count, and the hyperplane-subset count gives $t^{O(p+k)}$. Rational determinant bounds again control all bit lengths. This proves completeness without assuming a bound on the nullity of the full quadrilateral sector. □

Corollary 6.2 (Conditional recognition theorem)**.** *Suppose a procedure constructs and verifies, from every n-crossing knot diagram and in $n^{O(\log n)}$ bit time, a family of at most $n^{O(\log n)}$ exterior triangulations of polynomial encoded size. Suppose a fixed known constant C guarantees that every unknot has in one of these sources a standard vertex meridian with $p + k \le C\log(n+2)$. Then complete execution of the budgeted searches above gives unknot recognition in $n^{O(\log n)}$ bit time.*

Proof. Put $r(n) = \lfloor C\log(n+2)\rfloor$. Discard sources with $p > r(n)$; on each remaining source search only up to $k = r(n) - p$. At least one promised source remains for an unknot, and it produces a verified disc by the coverage assumption and Theorem 6.1. A nontrivial knot has incompressible torus boundary, so no verified disc is returned. Completing all searches therefore decides the problem. The size and parameter assumptions make each search quasi-polynomial; the allowed number of searches preserves that bound. Their source certificates and output certificates are included in the accounting. □

This corollary identifies a concrete sufficient structural theorem. The present work proves its search component, not its coverage hypothesis. In particular, arbitrary triangulations cannot simply be substituted for the controlled family: Section 9 gives an explicit linear-cancellation obstruction when multiple vertices are permitted.

7 An exact one-edge escape from the coherent family

The pinned source contains a certified counterexample to coherent-only normal-disc discovery. Start with the two-tetrahedron layered solid torus, then perform six legal 2–3 moves, choosing the following (tetrahedron, face) in the current labelling:

$$(1,2),(0,1),(2,2),(3,1),(3,2),(6,0).$$

The full source, intermediate face pairings, and move certificates are in `synthesis/data/coherent-obstruction-certificate.json` of the pinned repository. Its final triangulation T_* has eight tetrahedra, one vertex and ten edges. Its Regina isomorphism signature is

`iHvrQAkbgfeffhhasssvcaj`.

Legal Pachner moves preserve the underlying solid torus; the independently replayed source history is part of the evidence, not an inference from a successful disc query.

7.1 The coherent surface and the new vertex disc

The following table gives the primitive coherent surface C and the new meridian D. Coordinates in a row are $(t_0, t_1, t_2, t_3; q_{01|23}, q_{02|13}, q_{03|12})$. The semicolon separates triangle and quadrilateral coordinates.

Tetrahedron	Coherent C	Vertex disc D
0	$(2,2,0,0;0,0,2)$	$(2,2,0,0;0,0,2)$
1	$(0,0,2,2;0,2,0)$	$(0,0,2,2;0,2,0)$
2	$(0,0,1,2;0,0,2)$	$(0,0,1,2;0,0,2)$
3	$(0,2,0,2;0,0,1)$	$(0,2,0,2;0,0,1)$
4	$(2,0,0,2;0,1,0)$	$(2,0,0,2;0,1,0)$
5	$(0,1,0,1;3,0,0)$	$(2,3,0,1;1,0,0)$
6	$(0,1,2,0;2,0,0)$	$(2,3,2,0;0,0,0)$
7	$(0,1,0,1;2,0,0)$	$(2,3,0,1;0,0,0)$

Table 1: An exact 47-piece meridian outside the primitive coherent family. All coordinates are small integers; the accompanying certificates bind them to the full face-pairing source.

In increasing validator edge-root order, the weights are

Edge root	0	3	4	6	13	15	18	22	24	30
$w(C) = b$	6	2	4	2	3	1	3	5	3	1
$w(D)$	6	2	4	2	3	1	3	5	3	5

The boundary edge roots are $3, 13, 22$, so the changed edge 30 is interior. Its cancellation multiplicity is $z_{30} = (5-1)/2 = 2$. All other cancellation multiplicities vanish.

The exact certificates give C 41 pieces, Euler characteristic -3, one connected orientable component and one boundary curve. It has genus two. The new D has 47 pieces, Euler characteristic one and one essential disc component. Its standard support contains 25 positive coordinates; the supported matching matrix has rank 24. Thus the ray is a standard vertex ray, not only a ray made extreme by the selected weight space. The primitive cocycle lattice has rank one, so every primitive coherent surface comes from c or $-c$ and has exactly the same unsigned vector. A zero-cancellation meridian would therefore equal C, which is impossible. Consequently

$$\kappa_{\mathrm{v}}(T_*) = 1.$$

The previously retained 51-piece disc changes two edge weights, at roots 15 and 18, each by four. The new disc improves that explicit witness and uses fewer cancellation edges. It does *not* enlarge coverage beyond the existing complete search in the coherent surface's full quadrilateral sector: D belongs to that sector, with two of its allowed quadrilateral coordinates becoming zero. The new contribution is a complete parameterization and a less expensive way to discover the disc.

7.2 The whole two-dimensional cone is explicit

Consider the affine slice of $W_{\{30\}}$ in which the coherent coefficient is one:

$$w(s) = b + s\mathbf{e}_{30}.$$

Only tetrahedra 5, 6 and 7 change under the local reconstruction. In each, the coordinates t_0, t_1 increase by $s/2$; their occupied q_0 coordinates become respectively

$$3 - s/2, \quad 2 - s/2, \quad 2 - s/2.$$

All remaining coordinates are fixed. These vectors are admissible exactly for $0 \le s \le 4$. Negative s would make t_0 negative, while $s > 4$ fails the opposite-sum maximum condition in tetrahedra 6 and 7. Nonnegativity on edge 0 forces the coherent coefficient $\lambda \ge 0$. There are no admissible nonzero rays with coherent coefficient zero: putting weight only on edge 30 gives a negative face-corner expression. Thus the admissible cone in this weight space has just the two boundary rays C and D.

For $0 \le s \le 4$, direct cell counting gives

$$\chi(w(s)) = -3 + s, \qquad \#\text{pieces} = 41 + \frac{3s}{2}.$$

At $s = 0, 2, 4$ the vectors are integral. Independent full topology replay finds a connected orientable surface with one boundary curve in every case, of genus $2, 1, 0$, respectively. The geometry behind the search is therefore visible: the endpoint $s = 4$ removes the remaining handle obstruction while changing only one interior edge weight. The genus-one intermediate vector is not an extreme ray of this two-dimensional cone.

The complete, untruncated arrangement enumeration on this source gives:

Cancellation radius	Weight spaces	Intersections	Admissible rays	Disc vectors
0	1	1	1	0
1	11	73	2	1
2	56	5,634	5	2

All five rays found at radius two happen to be standard vertex rays. This observation is specific to the source; the general algorithm may emit rays that are extreme only in a restricted space. The counts here include complete enumeration and therefore differ from early-return search counters in the benchmark.

8 An infinite dense family with one cancellation edge

The following elementary extension lemma turns the certified eight-tetrahedron example into an infinite family. It also explains why large coordinates and quadrilaterals in almost every tetrahedron need not enlarge the new discovery parameter.

8.1 A support-rigid coherent extension

Let a tetrahedron $\Delta = 0123$ be attached along the faces 012 and 013, so that its old edges are all the edges except 23. For real heights h_0, h_1, h_2, h_3, the *coherent local vector* is obtained by taking the level sections of the affine height function. If $h_{v_0} \le h_{v_1} \le h_{v_2} \le h_{v_3}$, its potentially nonzero coordinates are the triangle at v_0, the quadrilateral separating $\{v_0, v_1\}$ from $\{v_2, v_3\}$, and the triangle at v_3, with coefficients respectively

$$h_{v_1} - h_{v_0}, \qquad h_{v_2} - h_{v_1}, \qquad h_{v_3} - h_{v_2}.$$

Tied heights cause zero coordinates; the statement includes those cases.

Lemma 8.1 (Injectivity on the coherent support)**.** *The five old edge weights determine a normal vector supported in the positive support of this coherent local vector uniquely.*

Proof. The supported coordinates are successive height gaps in the displayed fixed ordering. Their edge weights are signed height differences, with the signs fixed by that ordering. Knowing the weights on the connected graph $K_4 \setminus \{23\}$ determines the four heights up to a common additive constant, and hence determines every height gap. This argument is linear and remains valid when some of the gaps are constrained to zero. □

Proposition 8.2 (Preservation of a standard vertex ray)**.** *Suppose a standard vertex normal vector x on the old triangulation extends across Δ by adding a coherent local vector u. Then $x' = (x, u)$ is again a standard vertex vector after primitive normalization. If x is primitive integral and u integral, no normalization is needed.*

Proof. In any nonnegative matching decomposition $x' = y + z$, restriction to the old tetrahedra gives a decomposition of x. Its vertex property gives $y_{\text{old}} = ax$ and $z_{\text{old}} = (1 - a)x$ for some $0 \le a \le 1$. Matching on the attaching faces fixes the five old edge weights of y_Δ to be a times those of u. Nonnegativity forces y_Δ to be supported in the positive support of u. The preceding lemma gives $y_\Delta = au$. Thus $y = ax'$ and the ray is extreme. An integral extension of an integral vector with coordinate gcd one still has coordinate gcd one. □

8.2 Euler characteristic under a torus diagonal flip

Assume the attaching faces are the two triangles of a one-vertex torus, and that the local heights obey the parallelogram identity

$$h_0 + h_1 = h_2 + h_3. \tag{P}$$

This identity is the cocycle compatibility condition on the four sides of the lifted quadrilateral. Adding Δ performs the boundary diagonal flip and preserves the underlying solid torus.

For a finite set A of real heights, write $\operatorname{span}(A) = \max A - \min A$. The change in the normal surface Euler characteristic is

$$\begin{aligned}\Delta\chi = {} & |h_2 - h_3| - \operatorname{span}(h_0, h_2, h_3) - \operatorname{span}(h_1, h_2, h_3) \\ & + \operatorname{span}(h_0, h_1, h_2, h_3).\end{aligned} \tag{E}$$

Indeed, the new edge 23 contributes the first term to the normal surface's zero-cells. The two new boundary faces contribute the two subtracted arc counts. The coherent pieces in the new tetrahedron contribute the final term. All old cells retain their previous counts.

Lemma 8.3 (Zero Euler correction)**.** *Under (P), the right hand side of (E) is zero.*

Proof. Put $a = \min(h_2, h_3)$ and $b = \max(h_2, h_3)$. Since $h_0 + h_1 = a + b$, either both h_0, h_1 lie in $[a, b]$, or one lies $\delta \ge 0$ below a and the other lies δ above b. In the first case, all three spans in (E) equal $b - a$. In the second case, the two subtracted spans equal $b - a + \delta$ and the last equals $b - a + 2\delta$. Both cases give zero. □

If x is a primitive vertex meridian disc whose weights on the three boundary edges are coherent for the primitive cocycle, this extension is again a meridian disc. The extension is a primitive vertex surface, so it is connected: otherwise its distinct component vectors would decompose its ray, and primitivity excludes multiple copies. Its Euler characteristic is one by the lemma. Its new boundary is the coherent curve dual to the same primitive boundary cocycle, which is a

single essential curve on the two-triangle torus. Therefore the connected surface is an essential disc. Equivalently, the boundary statement follows by writing the two-triangle torus as an affine parallelogram quotient: a primitive integral linear functional has a connected essential level curve.

The coherent surface for the primitive cocycle also preserves its genus and number of components in this construction. Its coorientation extends across the tetrahedron. Every new coherent normal piece meets at least one attaching face: their two height intervals overlap on the height interval of edge 01 and their union is the complete height interval of Δ. Consequently a connected old coherent surface remains connected. The boundary remains one primitive coherent curve, and the Euler correction is zero.

8.3 The explicit Fibonacci family

Let $T_0 = T_*$ be the certified eight-tetrahedron solid torus from Section 7. Its primitive coherent surface C_0 has 41 normal pieces, Euler characteristic -3, and one boundary component, so it has genus two. Let D_0 be the new 47-piece standard vertex meridian given in the accompanying exact certificate. It differs from the absolute cocycle weights only at the interior edge numbered 30, where $w_{D_0}(30) = |c(30)| + 4 = 5$. Thus its cancellation support has exactly one edge. The boundary weights are $2, 3, 5$.

At each step layer a tetrahedron along the boundary edge of smallest weight. With $F_1 = F_2 = 1$, the jth added tetrahedron, $j \geq 1$, has coherent heights, after a local relabelling and perhaps reversing all signs,

$$(h_0, h_1, h_2, h_3) = (0, F_{j+2}, -F_{j+3}, F_{j+4}).$$

The identity $F_{j+4} = F_{j+3} + F_{j+2}$ gives (P). The old diagonal has weight F_{j+2} and the new diagonal has weight F_{j+5}. The two other boundary edge weights are unchanged. Hence the three boundary weights after r layers are

$$F_{r+3}, \quad F_{r+4}, \quad F_{r+5}.$$

The new local triangle coefficients are F_{j+3} at the height minimum and maximum, and its quadrilateral coefficient is F_{j+2}. It contributes $2F_{j+3} + F_{j+2} = F_{j+5}$ normal pieces.

Theorem 8.4 (Dense one-cancellation family)**.** *For every $r \geq 0$, the preceding construction gives a one-vertex solid-torus triangulation T_r with $t = r + 8$ tetrahedra and a primitive standard vertex meridian D_r such that:*

1. *its cancellation support is exactly the original interior edge 30;*
2. *its quadrilateral support has size $r + 6 = t - 2$;*
3. *its number of normal pieces is $F_{r+7} + 34$;*
4. *the primitive coherent surface C_r is connected, has one boundary component, genus two, and $F_{r+7} + 28$ pieces.*

In particular, dense quadrilateral support, exponentially large normal coordinates, and persistent failure of the coherent seed are compatible with a cancellation-support discovery parameter equal to one.

Proof. The base assertions are established by the supplied exact certificates and Table 1. All four vertices of each new tetrahedron belong to attaching faces, so the source retains one vertex. The old cocycle coordinates have gcd one and are unchanged, so the extended cocycle stays primitive. The support-injectivity proposition preserves the standard vertex ray, and the Euler and boundary arguments preserve the meridian disc and the genus-two coherent surface. All new coordinates are coherent for the extended cocycle; the old defect edge is interior and never changes. Thus the cancellation support stays exactly $\{30\}$. Since C_r remains a genus-two surface, no meridian has empty cancellation support; hence $\kappa_{\mathrm{v}}(T_r) = 1$. Each new tetrahedron has a positive quadrilateral

coefficient, so the support increases by one at each step. Finally,

$$\sum_{j=1}^{r} F_{j+5} = F_{r+7} - F_7 = F_{r+7} - 13.$$

Adding 47 or 41 yields the stated piece counts. The largest coordinate has $\Theta(r)$ bits even though its value grows exponentially. □

This theorem concerns the discovery parameter. The base disc lies in the same full quadrilateral sector as the coherent surface, so the family does not prove a coverage improvement over an existing complete search supplied with that sector. The new theorem keeps a low-dimensional edge-weight restriction through ray discovery, independently of the number of tetrahedra with quadrilaterals.

Exponential coordinate growth in layered solid tori is established in normal-surface complexity theory; Burton gives an explicit Fibonacci vertex-disc construction [7]. The distinguishing feature here is its persistence together with a genus-two coherent obstruction and exactly one cancellation edge. The table below gives some exact sizes; the saved benchmark also reaches $r = 64$.

Layers r	Tetrahedra	Disc pieces	Occupied quadrilateral types
0	8	47	6
1	9	55	7
2	10	68	8
4	12	123	10
8	16	644	14
16	24	28,691	22
32	40	63,246,020	38
64	72	308,061,521,170,163	70

9 A linear cancellation obstruction for arbitrary sources

A logarithmic cancellation bound cannot hold for all finite triangulations. The obstruction below uses multiple vertices; it is therefore separate from the one-vertex theorem and does not resolve its most interesting remaining geometric question.

Theorem 9.1 (Many forced cancellation edges)**.** *There are generalized triangulations U_r of a solid torus with $t_r = 6r + O(1)$ tetrahedra such that every oriented meridian disc transverse to their one-skeleton has at least r cancellation edges. In particular $\kappa_{\mathrm{v}}(U_r) \geq r = \Omega(t_r)$.*

Proof. Choose a fixed Whitehead curve J in a solid torus V. Its winding number is zero, while its geometric index is two: every meridian disc of V meets J at least twice. These properties are established, for example, by the geometric-index calculations in Section 5 and Figure 3(a) of Andrist–Garity–Repovš–Wright [8].

Choose a fixed tubular neighbourhood $N(J) \cong S^1 \times D^2$. Triangulate D^2 with three fixed boundary vertices and r interior vertices $p_1, \ldots, p_r$. Repeated insertion of an interior vertex into one triangle gives exactly $2r+1$ triangles and leaves the boundary triangulation unchanged. Put the one-vertex, one-edge Δ-complex structure on S^1. Give the disc vertices a global order, retaining the boundary order, and triangulate every product triangular prism by the compatible three-tetrahedron staircase subdivision. The resulting product has $3(2r + 1)$ tetrahedra. Each

$$e_j = S^1 \times \{p_j\}$$

is one global loop edge, and the r loop edges are distinct and disjoint. The triangulation induced on $\partial N(J)$ is independent of r.

Glue to a fixed triangulation of $V \setminus \operatorname{int} N(J)$. If their boundary triangulations need reconciliation, use a fixed triangulated collar. It does not touch the interior loop edges. The total tetrahedron count is $6r + O(1)$ and the underlying manifold is the solid torus V.

Every e_j is parallel to J in its fixed neighbourhood, hence has winding number zero and geometric index two in V. For every oriented meridian D,

$$c_D(e_j) = 0, \qquad w_D(e_j) \geq 2.$$

Equation (4) makes each e_j a cancellation edge. This argument does not depend on a choice of vertex potential: every coboundary evaluates to zero on a loop edge. There are r such edges, which proves the bound for all meridians and hence for vertex meridians. □

The theorem is a representation obstruction, not a lower bound for unknot recognition. Its source has $\Theta(r)$ vertices, and the theorem does not prove that these obstructions survive a controlled reduction to one vertex. The product construction deliberately makes the loops separate vertices so that their triangulation cost and cancellation contributions are transparent.

A successful global approach must therefore choose its sources. It may need certified retriangulations that remove such repeated gadgets, a parameter measuring independent obstructions rather than literal edge support, or a family of alternative exterior presentations. A bound on one convenient triangulation chosen by an unspecified topological existence argument would still be insufficient: both discovery of that source and the cost of certifying its relation to the diagram belong in the complexity theorem.

10 Ordered interfaces and safe disc-bundle replacement

The maintained source parallelity packet already describes a cut as a bounded inherited core together with interval bundles whose bases are binary families of polygons. It retains all selected 0-, 1- and 2-handles, including disc bases supplied only by a selected 2-handle. Its component profiles record Euler characteristic, orientability, boundary count and core-facing ports, but do not extract the ports' cyclic boundary order. Walking the subdivided boundary can take time proportional to normal weight. We supply an executable extraction of the finite interface and distinguish the correct replacements for fully and partially attached disc bundles.

These are additions to this implementation, not claims of a new general parallelity algorithm. Lackenby's Theorems 9.2 and 9.3 already provide polynomial algorithms with attachment data, including cyclic order around vertical annuli [10]. Report 68 in the repository already proves the degree-two ordered-boundary principle, recovering cut-path endpoints from their count, sum, square sum and path length. Our contribution is its integration with the later all-dimensional source packet, endpoint occurrences at coincident vertices, zero-gap flags, fibre transport, correct partial-disc replacement, and independently replayable code.

10.1 Extracting the port matching

Let B be a finite compact surface represented by s interval records of polygon copies and affine side gluings. There are N corner addresses. Let V denote the corner-pairing system induced by glued polygon sides; its orbits are quotient vertices. Distinguish p individual free sides, the *ports*, each specified by a binary rank. Port j has two endpoint occurrences $e_{j,0}, e_{j,1}$. These labels remain distinct when their addresses represent the same vertex. Other free sides are old-boundary sides. Let R be V together with the endpoint pairings of old-boundary sides, omitting the endpoint pairings of ports.

Theorem 10.1 (Finite interface extraction)**.** *Each R-orbit containing marked endpoints contains exactly two endpoint occurrences. Their pairing τ, and the information whether each paired gap has zero length, determine all cyclic port words. They are computable using $O(p^2)$ ordinary interval-orbit counting calls on systems with $O(s+p)$ records and universe size N. Hence an algorithm exists whose running time is polynomial in $s+p+\log(N+1)$; the output has $O(p)$ combinatorial records.*

Proof. After quotienting by V, the free sides cellulate the disjoint union of circles ∂B. Delete the open port sides. Each remaining component is a circle with no marked endpoint, or an interval with two marked endpoint occurrences. A degenerate interval may be a single vertex, as when ports are adjacent or a single loop edge is deleted. The R-orbits describe exactly these components. Thus the marked orbits give a perfect matching τ of the endpoint occurrences.

For any pairing system Q, let $c(Q)$ be its orbit count. Adding one singleton pairing gives

$$c(Q \cup \{x \leftrightarrow y\}) = \begin{cases} c(Q), & x \sim_Q y, \\ c(Q) - 1, & x \not\sim_Q y. \end{cases}$$

One edge merges precisely two previously distinct components, or none. Testing all pairs of marked addresses therefore recovers τ using $O(p^2)$ calls. For a matched pair, the analogous test in V detects a zero-length gap: its ends represent the same vertex exactly when the remaining interval contains no old-boundary side. A nondegenerate interval cannot have its ends identified in a cellulation of a one-manifold.

Let σ exchange the two occurrences belonging to each port. The multigraph with the σ and τ pairs as edges is 2-regular. Its alternating cycles are exactly the boundary circles containing ports. Traversing them gives the cyclic words in $O(p)$ time after sorting labels; choose the least port label and its endpoint zero to fix each direction. The polynomial bound follows from the AHT orbit-counting theorem [3]. When $p = 0$ the answer is empty and no counting call is needed. □

For the maintained source packet, $s = O(T)$, $p \le 66T$, and $\log(N+1) = O(L + \log(T+1))$, where T is the tetrahedron count and L bounds the coordinate bit length. The theorem therefore yields a polynomial reduction in encoded source size. This does not identify the repository's particular interval implementation with the proved AHT bound: its currently established replay bound is in the realized trace size.

The prototype uses two marked weighted queries in place of repeated orbit counts. Endpoint $e_{j,a}$ receives weight 2^{2j+a} at its corner address; overlapping weights are added. Positive weights of R-orbits have exactly two set bits, and positive weights of V-orbits have one or two. Each endpoint occurs once, so no bit carries arise and every positive mask has multiplicity one. All unmarked orbits share the zero-weight row. The weights have $2p$ bits, and both positive histograms have $O(p)$ rows.

10.2 Fibre transport and the attachment subset

Each polygon gluing has a fibre-reversal bit ϵ. Lift its corner map to two sheets by

$$(x, a) \longmapsto (g(x), a \oplus \epsilon).$$

Old-boundary side maps preserve the fibre in their polygon charts. Applying the matching construction to the lifted system gives the fibre change along each gap, by following sheet zero at one end to its sheet at the other. This is the prototype's third weighted query, using $4p$ endpoint bits.

Proposition 10.2 (Signed consistency and attachment components)**.** *The lifted matching commutes with the deck involution. For an orientable total interval bundle, the xor of gap fibre changes around every marked boundary circle is zero. Let $A \subseteq \partial B$ be the union of port sides. A recovered circle lies entirely in A exactly when every gap has zero length. On any other recovered circle, the number of components of A is the number of nonzero gaps.*

Proof. An interval bundle over an interval or a point is trivial, so each gap has two distinct lifts exchanged by the deck involution. In an orientable total space, the fibre orientation character equals that of the base. The restriction of the tangent bundle of the base to its boundary splits as the tangent line of a circle and an outward normal line, both orientable. Thus fibre holonomy is zero on every boundary circle, including boundaries of nonorientable bases. Port traversal itself preserves the local fibre, giving the xor assertion. Finally, delete the nonzero gaps in a circular word: either none is deleted and A is the full circle, or each such gap separates one of its interval components from the next. □

This does not imply that the full bundle is a product: an orientable interval bundle over a Möbius band has nontrivial fibre holonomy on its core curve and trivial holonomy on its boundary. Nor do independently directed cyclic words supply coherent orientations between different boundary circles or arbitrary markings on horizontal boundaries. Those data must be carried separately when a patterned hierarchy requires them. If a base profile has b boundary circles and c recovered marked circles, its unmarked circles remain the binary count $b - c$.

10.3 Disc products with full and partial attachments

Let a disc product $E = D^2 \times I$ meet the complementary core K in $A \times I$, with its source attachment charts, where A is a union of port sides in ∂D^2. The interfaces are embedded, and interfaces of different bundle components are disjoint. These are hypotheses supplied by the authenticated source packet.

Theorem 10.3 (Safe local replacement)**.** *If $A = \varnothing$, E is an isolated 3-ball. If $A = \partial D^2$, adjoining E attaches one 2-handle. Otherwise let $r \geq 1$ be the number of interval components of A. Adjoining E is, after transport through collars of the attaching discs, the addition of $r - 1$ relative 1-handles. The homeomorphism can be fixed outside those collars and transports all explicitly retained boundary markings. Over all attached disc bases, the number of relative handles added is at most p.*

Proof. The empty and full-boundary cases follow directly from the product description. In the remaining case, choose $r - 1$ disjoint properly embedded arcs in D^2 with endpoints outside A, cutting it into r discs, each containing precisely one interval of A on its boundary. Such arcs are constructed by successively cutting off the intervals in cyclic order; their dual graph is a tree. Their products with I are vertical discs splitting E into r balls. Each ball meets K in one disc and can be absorbed into a collar of that disc. The enlarged core is homeomorphic to K by a homeomorphism fixed away from the attaching collars. Rejoining across the $r - 1$ cut discs amounts to attaching $r - 1$ 1-handles between their collar patches. The same charts transport boundary markings.

A full attachment uses at least one port and contributes one handle. A partial attachment using p_i ports has $r \leq p_i$ and contributes $r - 1 \leq p_i$ handles. Sum over components. Unattached balls can occur with large multiplicity and must be counted rather than listed. □

The Euler contribution is $1 - \chi(A)$: it is 1 for an empty or full attachment, and $1 - r$ for a proper nonempty attachment. Thus a single proper interval gives a collar addition and no relative handle;

two intervals give one 1-handle. Replacing every disc product by a 2-handle would fail even this Euler check.

This theorem supplies local relative handles, not automatically an admissible handle structure of uniform type. It need not fix the entire originally embedded core pointwise, because an attaching boundary disc becomes interior during collar absorption. Constructing the transported pattern, arranging the required handle structure, and proving its admissibility and global hierarchy bounds remain distinct tasks [4].

10.4 Implementation evidence and next steps

The supplement contains a producer, an independent query constructor and certificate checker, a finite oracle that actually walks polygon boundaries, and a disc-replacement plan. Source geometry is authenticated upstream by `verify_normal_cut_parallelity`. The extra checker does not call the producer's query builder, decoder or traversal. Shared discovery caps return no interface on exhaustion; verification caps cover all three traces. Cancellation propagates.

The saved audit passes 11 test methods with 184 literal geometric comparisons, including 54 authenticated source comparisons. It checks all 126 markings of discs with one through six sides, annular and Möbius charts, source relabellings, rich 3-manifold fixtures, coincident endpoint addresses, and forged cyclic, gap, fibre and weight data. Exact JSON replay succeeds for 2^{8192} parallel meridians in four tetrahedra with producer routines disabled; the finite oracle refuses that input by its expansion cap.

Two complete example certificates accompany the audit. One source, `fibonacci_lst_02`, has two tetrahedra. Its coordinate rows are $(0, 0, 1, 1, 1, 0, 0)$ and $(1, 1, 1, 1, 0, 0, 0)$. It has one disc base supplied solely by a selected 2-handle. All eight ports lie consecutively around its boundary, giving a full-circle 2-handle attachment. Three parallel meridians in the two-tetrahedron layered torus instead have two attached disc bases with one proper core interval each: both are collar additions, and two further ball products are unattached. The prototype records these local plans but performs no surgery.

The next concrete objectives are to certify the induced marked boundary homeomorphisms; carry compressed pattern endpoints through these charts; convert the relative additions to admissible uniform-type handles with a proved size bound; and preserve coherent fibre data between different boundary circles for non-disc bases. The ordered interface removes an explicit reconstruction gap, while these remaining obligations prevent inferring a quasi-polynomial recognizer from this local operator alone.

11 Implementation, evidence and integration

The contribution is packaged against revision `90e16db798c8d976bbe0c062e07cf20565e0b63a`. The new production-facing files are additive: an edge-weight discovery module, focused tests, and a reproduction driver. The ordered-interface prototype and the polynomial vertex gate are separate supplements. No existing recognizer default or diagram-to-exterior conversion is changed.

11.1 Search implementation and contracts

The file `fast/fastunknot/normal_weight_defect.py` provides:

`local_coordinates_from_weights`
: Exact rational reconstruction using the four-point and face-corner tests. It permits rational rays, normalizing only after assembling standard coordinates.

`discover_in_weight_subspace`
: Complete arrangement search in a supplied W_F. It drops the coherent basis column precisely when b is supported in F, and deduplicates weight directions without leaving that space.

`weight_defect_search`
: The elementary general-k algorithm of Theorem 4.2. It visits all edge subsets of sizes zero through k, with global intersection and subspace budgets.

`one_edge_defect_search`
: The sharper sparse-incidence method for $k = 1$. It prepares the global hyperplane list once and evaluates Equation (12) at each row-edge incidence. The general-k row-matching improvement in Theorem 5.1 is proved but is not implemented by this routine.

For an actual one-edge meridian the offset is a positive even integer. The optimized method rejects other offsets, deduplicates the survivors, and reconstructs only tetrahedra incident to the changed edge. If f_e counts occurrences of e in one representative of each global face, the Euler correction is

$$\Delta\chi = \delta(1 - f_e/2) + \sum_{\Delta \ni e} (\#\text{new pieces in } \Delta - \#\text{old pieces in } \Delta).$$

This is the normal-surface cell count: geometric edge weights contribute vertices, half the edge sum of each global face contributes normal arcs, and the normal coordinates contribute faces. Repeated local occurrences of a global edge must be counted in f_e.

After $O(t)$ sparse preparation there are $O(t)$ arithmetic incidence attempts. Local reconstruction work is bounded by $O(\sum_e \deg(e)^2)$ in the worst case, since rows incident to an edge and tetrahedra in its star are both controlled by its degree. This may be quadratic in t. It also excludes the cost of assembling a successful full vector and verifying it. A linear candidate count is therefore not presented as a linear bound for the complete recognition task.

Every positive result includes a source-bound normal-disc certificate and passes independent replay before return. A restricted exhaustion status asserts only absence of a standard vertex disc in the searched class. There is no independently certified negative-search transcript. Exhausted resource budgets return `INCONCLUSIVE`, and caller cancellation propagates. The input triangulation is not modified. The total-Euler prefilter retains the promised primitive vertex disc, whose Euler characteristic is one; it is not a complete test of all disconnected vectors containing some disc.

The separate `vertex_gate.py` realizes Proposition 4.4. Its exact rank and boundary-coboundary checks supply a polynomial verification alternative for the promised vertex rays. It is kept separate from the timed discovery implementation so that the saved benchmark corresponds exactly to the source hashes in its result file. This distinction is also useful for integration: one may preserve richer disc-component discovery while exposing a narrower provably polynomial verification mode.

11.2 Controlled comparison on the layered family

Each fixture was given one excluded warm-up per arm followed by seven measured rounds. A fixed random seed shuffled arm order. Every positive time includes its own cocycle preparation, candidate construction, native disc observer and internal independent replay. A further fresh replay was timed separately. The retained timing run did not overlap this task's heavy audits. The environment was CPython 3.12.14 on x86_64 Linux; the artifact records the interpreter and source hashes.

The supplied-edge arm includes preparation of its two-dimensional arrangement, explaining why it is not necessarily faster than the globally prepared sparse-incidence arm. Coherent-only search completes quickly but finds no disc, so it is not included as a successful recognition-time comparator.

On all four labelled fixtures, the global method begins 39 bases, including the coherent attempt, and makes ten local tetrahedron reconstructions before its one positive disc query. The deduplicated

t	Seeded Q-sector	All W_F, $k = 1$	Global one-edge	Supplied edge	Ratio
8	0.003947	0.022281	0.002839	0.004663	1.39
16	0.023683	0.038538	0.005050	0.008571	4.69
40	0.354871	0.087642	0.012730	0.020739	27.9
72	2.114322	0.164222	0.023326	0.044324	90.6

Table 2: Median complete search time in seconds. Ratio is the existing seeded Q-sector median divided by the global one-edge median. All four positive methods return exactly the same vector on each source.

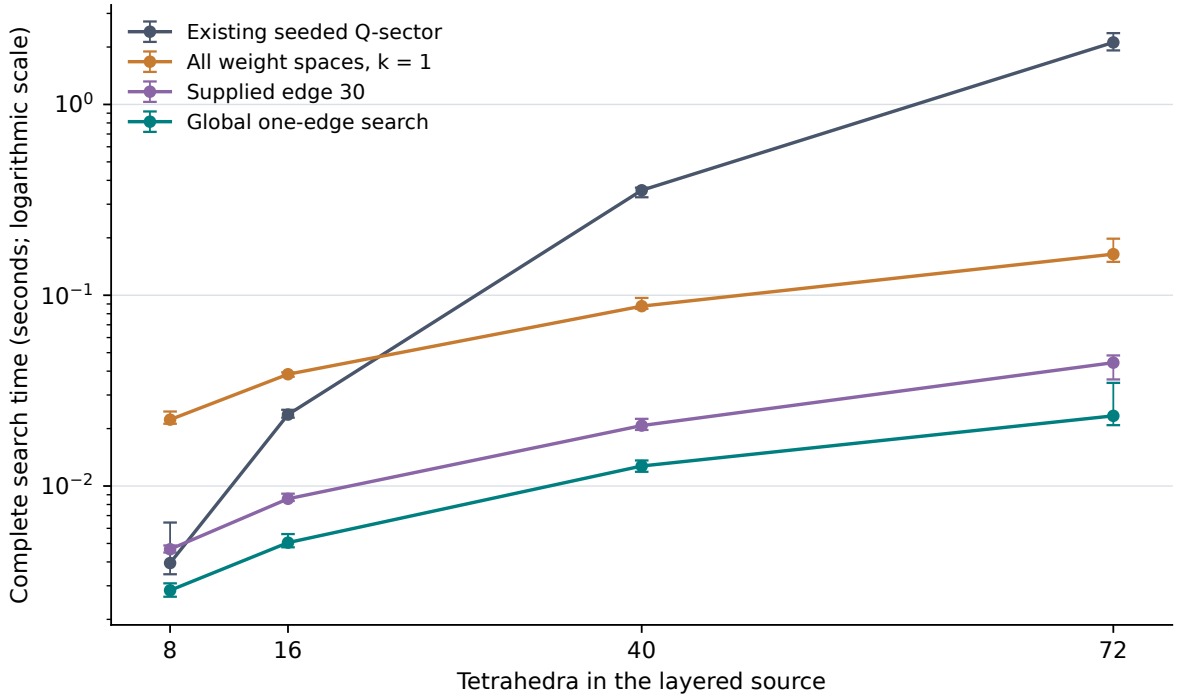


Figure 1: Measured medians and full observed ranges on four structured sources. The vertical axis is logarithmic. The lines connect measurements; they are not fitted asymptotic laws. Source data and the plotting script are supplied.

global hyperplane counts are 72, 128, 296 and 520. These early-return counters depend on the serialized edge ordering; the parameter and completeness statement do not.

The existing seeded sector's matching nullity remains three throughout the experiment. Its number of kernel variables grows through 16, 40, 112 and 208. The measured difference therefore reflects the amount and organization of exact algebra and setup, rather than a growing sector nullity. The elementary new all-subspace method is slower than the existing solver on the two smallest sources; that unfavorable comparison is retained. One family and four sizes do not establish a corpus-wide speedup. The asymptotic theorem comes from the proofs above.

11.3 Independent and adversarial verification

The focused edge-discovery suite has 13 tests. It covers rational local reconstruction, large binary coordinates, dependent coherent columns, the old and new explicit discs, global caps, cancellation, input immutability, and replay with corresponding producer functions disabled.

The polynomial vertex gate has seven supplementary tests, including multiple boundary vertices, rejection of a vertex link with nonzero raw parity but trivial cohomology class, nonextreme mixed surfaces, determinant checks against direct small permutation sums, and forged certificates. At 72

tetrahedra its supported minor has order 216. The compact certificate has 2,789 bytes, excluding the separately supplied source and normal vector; exact independent replay succeeds without an orbit query.

Regina independently enumerated 138 standard vertex surfaces across eight triangulations of one through eight tetrahedra. Exact edge reconstruction agrees on every vector. Twelve meridians satisfy the signed/unsigned parity lower bound; eleven applicable subspace searches recover their targeted standard vertex discs with cancellation support at most two. The one-edge optimized search also succeeds whenever its promise applies.

External topology checks confirm the layered family at $r = 0, 1, 2, 4, 8$. They are deliberately bounded by 10,000 normal pieces: Regina's installed documentation states that the connectivity and orientability methods used expand normal discs, and its compression test may cut and retriangulate. An initial unrestricted large-family external audit terminated with exit code 137; no conclusions use its unretained partial results. The saved record explicitly marks omitted external checks. Exact native binary certificates and supported-rank checks remain in force at larger sizes, including the 72-tetrahedron benchmark witness with more than $3 \cdot 10^{14}$ normal pieces.

The parallelity supplement's independent audit is described in Section 10; it adds 184 literal geometric comparisons, including 54 authenticated source comparisons, and huge binary-coordinate replay. These tests measure different obligations from vertex-disc discovery and are not added to its sample count.

The maintained suite contains 1,627 collected tests. Successful coverage was reconciled across the initial broad run (1,512 successful tests) and a 115-test targeted rerun after restoring 20 historical dependency files from the pinned revision. All 42 initial errors were missing-dependency or setup errors, with no assertion failures; none remains unresolved. Both logs, the restored-file list and the complete test identifiers are included. The 13 new discovery tests are included in the maintained count; the seven vertex-gate and eleven interface tests are additional.

These are exact-arithmetic programs and human-readable mathematical proofs with independent review. They are not proof-assistant formalizations. The principal residual mathematical obligation is the source-coverage theorem, while the principal remaining implementation obligations concern optimized general-k enumeration and the transport of boundary markings through actual geometric replacements.

11.4 Package and integration procedure

The archive contains the complete LaTeX sources, compiled PDF, figures, measurements, certificates, additive source overlay, a patch against the pinned revision, and reproduction instructions. A small baseline snapshot includes the dependencies needed by the focused research experiments; it is not a replacement for the whole repository.

To integrate, first apply the additive discovery files to the pinned source and run the focused suite. The interface and vertex-gate supplements can be reviewed independently. Preserve the three trust boundaries: a certified exterior source, a verified normal surface in that source, and a justified coverage statement before turning finite exhaustion into a global negative verdict. Broader default scheduling should await a mixed-source benchmark that includes unfavorable cases, relabellings, large-degree edges and preprocessing costs.

12 A research programme toward quasi-polynomial recognition

The new parameter moves the main question from discovering all local quadrilateral choices to controlling the places where geometric and algebraic intersections disagree. This is useful because a cohomology generator is inexpensive to compute, and the disagreement is a directly testable

geometric object. The next results should establish control of that object under certified changes of representation.

12.1 The structural theorem that would complete this route

The most direct objective is the following concrete question.

> Is there an explicit procedure, taking $n^{O(\log n)}$ bit time, which produces certified polynomial-size exteriors of an n-crossing diagram, such that an unknot has a standard vertex meridian with $p + |F(D)| \leq C \log(n + 2)$ in at least one produced source, for an absolute known constant C?

An affirmative answer combines with Corollary 6.2; nothing else in that corollary requires expanding a normal surface. Several aspects of the wording are essential. The favourable source must be found within the bound, its relation to the original diagram must be certified, and vertex potentials count in p. A construction with many vertices cannot silently invoke the one-vertex cancellation theorem. A promising normal disc that is not a standard vertex disc does not yet meet the promise.

This question is deliberately not elevated to a conjecture that all sources satisfy. The Whitehead construction rules out that broader statement. Even the restriction to all one-vertex triangulations remains an open problem in this investigation and may be false. Constructively choosing a family of representations is a more flexible target.

12.2 Geometric questions, in order of dependence

G1. **Can low cancellation be transferred to a vertex disc?** Suppose some essential normal disc has cancellation on at most k edges. Is there a standard vertex meridian with at most $f(k)$ such edges, ideally $f(k) = O(k)$, after a controlled source modification? The ordinary vertex-disc theorem does not include this support control. Intersecting a sector with W_F produces extreme rays of the intersection, which need not be extreme in the full sector. This is exactly the gap a proof must bridge. A useful first result would identify sufficient conditions on oriented normal sums under which no cancellation is created on an edge that was previously coherent.

G2. **Can the Whitehead obstruction persist with one vertex?** Construct a one-vertex family with $\kappa_{\mathrm{v}} = \Omega(t)$, or prove a quantitative obstruction to collapsing the many interior loop vertices in Theorem 9.1 without destroying the independent geometric-index barriers. Either result would substantially narrow the correct global claim. The geometric index of an individual embedded loop does not by itself control a whole collapsed graph, whose edges may share vertices and change their isotopy classes.

G3. **How does cancellation behave under certified moves?** For 2–3, 3–2, boundary layering and crushing operations, derive local inequalities for the best achievable cancellation support of an essential disc, while also controlling standard extremality. Boundary layering has the exact favourable theorem proved here. General Pachner moves can change the coherent baseline, so comparing only geometric weights before and after a move is insufficient. A useful move record should transport the cocycle, candidate disc, affected edge stars and source correspondence together.

G4. **Can repeated obstacles be removed in batches?** The Whitehead example has many parallel copies of a single geometric gadget. Develop a compressed quotient of repeated edge-star regions, with binary multiplicities and a proof that essential-disc discovery is preserved. A parameter counting independent gadget types might be small where literal cancellation support is linear. The hard part is retaining the ordered attachments and their fibre data;

merely aggregating edge weights can change the embedding problem. The interface operator of Section 10 supplies part of the data needed for a safe batch transformation.

G5. **Can a complexity potential force logarithmic depth?** Seek a certified simplification or hierarchy move that either exposes a meridian or decreases a nonnegative structural potential by a fixed factor. The potential should account for core size, boundary pattern, vertex-gauge dimension and independent cancellation regions. A decrease in coordinate weight alone is unsuitable: the layered family has huge weights but a constant discovery parameter. A decrease in tetrahedron count alone may fail during useful Pachner exploration. A potential theorem must bound branching as well as depth.

12.3 Algebraic and algorithmic questions

A1. **Implement the general sparse-incidence theorem.** The highest-confidence next coding task is the row-first enumeration in Theorem 5.1, including its exceptional normalization case. Use partial injective row-to-edge matchings, incremental exact rank and canonical descriptions to avoid permutations of the same system. Completeness should first be compared with the elementary algorithm at $k = 2, 3$ on small sources. No heuristic pruning should be credited in the asymptotic bound without a certificate that it preserves every relevant invertible system.

A2. **Can the exponent depend on interaction rather than support?** The determining equations form a hypergraph with row size at most four. Study dynamic programming over certified decompositions of the affected edge-star interaction graph. A bag must retain compatible four-point sectors, rational boundary weight relations and enough global information to verify a primitive essential disc. Storing only possible total Euler characteristics would be unsound. The goal is an explicit bound in both the interaction width and the coefficient bit length, followed by a geometric theorem controlling that width on selected sources.

A3. **Can algebraic signatures reduce duplicate active systems?** The present global list has many repeated hyperplanes, and one ray may have many active descriptions. Investigate reverse search on active bases or rank certificates indexed by sparse minors. A useful target is polynomial delay per distinct candidate on a fixed W_F. The number of distinct candidates can still be exponential in the dimension, so such an output-sensitive result would improve implementation without alone settling the global complexity question.

A4. **Can the vertex gate replace more expensive positive checks?** The exact minor and boundary parity certificate is complete for standard vertex meridians. Measure its crossover with the existing exposed-unit ray gate and compressed component observer. A sensible schedule first uses cheap sufficient rank certificates, then the complete supported rank test when a theoretical vertex-only mode is requested, and reserves the richer observer for additional restricted rays. Source preparation and reuse of matrix factors must be included in comparisons.

A5. **How can multiple-vertex sign chambers be compressed?** The current extension pays $t^{O(p)}$ for the arrangement of cocycle hyperplanes. Examine whether a certified spanning-tree gauge, boundary collapses or a small set of relevant vertex potential differences can reduce p without excluding the unknown disc. Fixing arbitrary vertex potentials is not harmless: signed intersections of the actual disc may use a different representative. A valid reduction must transport the geometric problem, rather than simply delete gauge variables algebraically.

12.4 Compressed geometry and hierarchy questions

H1. **Construct the boundary homeomorphisms, not only handle plans.** For each full or partial disc-base attachment, output the actual collar maps and transported marked arcs. Prove that disjoint source interfaces remain disjoint and that the output records their induced order. The correct relative handle count is already known here; an executable topological replacement requires more data than that count.

H2. **Convert the result to admissible handles.** Show that the transported relative additions can be changed to the uniform-type handle structures required by the hierarchy theory, with polynomial encoded size. State precisely which boundary patterns are preserved and which tests establish admissibility. A bounded inherited core plus interval bundles does not automatically provide this step.

H3. **Extend fibre information beyond separate boundary circles.** The present signed double cover records transport around each marked boundary circle. Non-disc bases may require paths connecting distinct circles and coherent horizontal boundary markings. Identify a finite set of such paths, bound its size, and certify its relation to the source charts without expanding the base. Annular and Möbius examples should remain explicit regression cases.

H4. **Close the orbit-engine complexity gap.** The interface theorem has a polynomial reduction to ordinary AHT counting; the implementation uses weighted traces. Either prove that these concrete traces and their arithmetic stay polynomial in encoded source size, or substitute a rigorously bounded counting implementation for the proof-critical mode. Fast behaviour on long binary multiplicities is valuable evidence, but does not control every possible interval system.

12.5 An experimental programme with useful failure criteria

The next corpus should combine several independently generated classes: small enumerated solid tori, diagram exteriors with verified provenance, Pachner walks, layered sources, and sources containing controlled geometric-index gadgets. For each, record t, vertex count, cocycle rank, the best certified vertex cancellation support found, sector nullity, maximum coordinate bit length and edge-star degree distribution. Keep a distinct field for a lower bound or exhaustive result; a heuristic failure to find a disc is not a lower bound on κ_{v}.

Compare equal workloads: the same supplied source, the same necessary preparation, the same certificate acceptance contract, and separately reported fresh replay. Relabel every source several times to measure early-return ordering effects. Record failures and cap exhaustion beside successes. In particular, the elementary all-subspace method's slower small-source performance should remain visible when evaluating the optimized replacement.

Three falsifiable milestones would make the next iteration decisive:

1. General-k row-first enumeration exactly matches elementary enumeration on a complete small corpus while reducing actual attempted systems and elapsed time.
2. A certified move family has a proved inequality for cancellation support or gauge dimension and preserves source size within an explicit bound. A single verified counterfamily should trigger revision of the proposed structural claim.
3. Partial disc-base replacements output independently checked boundary maps and an admissible next handle structure on both the full-circle and collar fixtures, before being used inside a hierarchy.

13 Conclusion

Sparse edge cancellation gives a complete way to discover standard vertex meridians without first supplying their quadrilateral sector. Its search spaces have dimension at most $k+1$ in the one-vertex rank-one case, and sparse equation support gives the sharper explicit enumeration bound. Exact rank, Euler characteristic and boundary parity suffice to verify every promised vertex disc in polynomial bit time. The new disc and its infinite layered extensions demonstrate that this parameter can remain constant in the presence of dense quadrilateral support, exponential normal weights and persistent failure of coherent-only discovery.

The Whitehead construction shows why source selection is an essential part of the proposed global

theorem. Ordered parallelity interfaces provide an additional geometric operation needed for controlled changes of source, but the admissible replacement and coverage arguments still require proofs. The precise remaining objective is therefore a constructive, certified source family with logarithmic $p + k$. Proving it, or finding the right replacement parameter when it fails, is the central research task exposed by this continuation.

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